

Complete Data Preparation Guide

Step 1 — Define the Goal and Collect Sample Data

- Define a one-sentence objective.
- Identify data formats (CSV, Excel, JSON, SQL dump).
- Gather small sample data (100–1000 rows).
- Write down known issues (missing values, duplicates, wrong formats).
- Decide desired output (clean CSV, Parquet, or SQL table).

Step 2 — Set Up VS Code Environment

1. Install Python 3.9+.
2. Create a project folder.
3. Open it in VS Code.
4. Create a virtual environment:

```
python -m venv venv
```

5. Activate it:

```
venv\Scripts\activate
```

6. Install libraries:

```
pip install pandas numpy openpyxl pyarrow jupyter
```

7. Create a test script to ensure everything is working.

Step 3 — Load and Inspect Data

```
import pandas as pd
```

```
df = pd.read_csv("data.csv")
```

```
df.head()
```

```
df.info()
```

```
df.describe(include="all")
```

```
df.duplicated().sum()
```

Step 4 — Clean the Data

Handling missing values:

```
df.fillna("Unknown")
```

Removing duplicates:

```
df = df.drop_duplicates()
```

Cleaning text:

```
df["name"] = df["name"].str.strip().str.lower()
```

Fixing dates:

```
df["date"] = pd.to_datetime(df["date"], errors="coerce")
```

Fixing numeric columns:

```
df["price"] = pd.to_numeric(df["price"], errors="coerce")
```

Step 5 — Transform the Data

Renaming:

```
df.rename(columns={"old": "new"}, inplace=True)
```

Computed columns:

```
df["total"] = df["qty"] * df["price"]
```

Encoding categories:

```
df["category_code"] = df["category"].astype("category").cat.codes
```

Step 6 — Combine and Merge Datasets

Concatenate:

```
df_all = pd.concat([df1, df2, df3])
```

Merge:

```
merged = df.merge(other_df, on="id", how="left")
```

Step 7 — Validate the Cleaned Dataset

```
df.dtypes
```

```
df.isna().sum()
```

Step 8 — Export the Clean Data

```
df.to_csv("clean_data.csv", index=False)
```

```
df.to_excel("clean_data.xlsx", index=False)
```

```
df.to_parquet("clean_data.parquet")
```

Step 9 — Automate the Workflow

Example clean_data.py:

```
import pandas as pd

def clean():
    df = pd.read_csv("raw.csv")
    df = df.drop_duplicates()
    df["name"] = df["name"].str.strip()
    df.to_csv("clean.csv", index=False)

if __name__ == "__main__":
    clean()
```

Example Project Structure

```
data-prep-project/
raw/
cleaned/
scripts/
notebooks/
venv/
```